UNIT – 5 (Part-1)

Getting Started with Linux, Linux Booting, Linux Admin (Ubuntu)

Topic: 1 Give the introduction of Linux history. Mark 2

Ans.

Detailed:

- Linux started in 1991.
- Linux was created by Linus Torvalds. Initially UNIX was available as operating system, but it was used on huge computers with business and universities in 1970.
- Linus had tried to develop one program especially for the PC with an 80386 processor and independent of any operating system.
  He was developing with use of MINIX (MINIX is kind of operating system which is clone of UNIX.

Topic: 2 Explain GNU and GPL concept. Mark 2

Ans.

Detailed:

- To prepare free UNIX like operating system, Richard Stallman has started a project called GNU General Public License (GPL).
- In 1990 enough software were available to create full operating system then also GNU was not successful.
- The GNU software was distributed for free to anyone who wanted the source code was also available so user can modify the code according to their need.
- GNU and Linux developers worked to integrate GNU components with Linux kernel to make full functional and free operating system.

Topic: 3 Explain Open source and freeware. Mark 2

Ans.

Detailed:

- Open source means the software code is available with licensed in which the copyright holder provides the rights to study, modify, and redistribute the software to anyone without any cost and with any purpose.

- Freeware means software is available without cost.

- Freeware generally available with some restricted rights.

- There is not proper definition for freeware and open source, but it can be identified in following terms.

- Anyone can join the open-source project, where in freeware a closed development group is developing the software.
- In open-source code is available so new version can be created from the existing software, where in free ware code is private.

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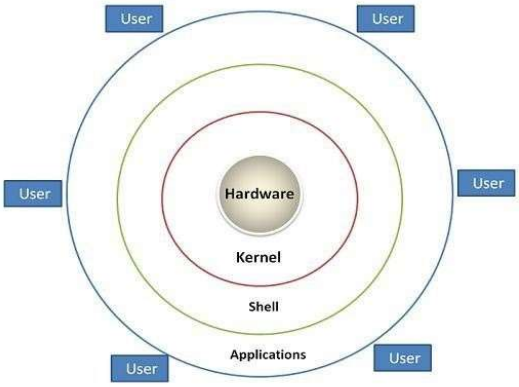
- **Topic: 4 Explain structure & features of Linux.** **Mark 5 Ans.**



Linux System Architecture

Detailed:

Linux Architecture

- The Linux system works basically on 4 layers.

- First view the diagram above, which is showing the architecture of a Linux System.
- **Hardware –** Hardware consists of all physical devices attached to the System. For example: Hard disk drive, RAM, Motherboard, CPU etc.
- **Kernel –** Kernel is the core component for any (Linux) operating system which directly interacts with the hardware.
- **Shell –** Shell is the interface which takes input from Users and sends instructions to the Kernel, Also takes the output from Kernel and send the result back to output shell.
- **Applications –** These are the utility programs which runs on Shell. This can be any application like Your web browser, media player, text editor etc.

Linux Features

- Following are some of the important features of Linux Operating System.

- **Portable** – Portability means software's can works on different types of hardware's in same way. Linux kernel and application programs supports their installation on any kind of hardware platform.
- **Open Source** – Linux source code is freely available, and it is community-based development project. Multiple teams work in collaboration to enhance the capability of Linux operating system and it is continuously evolving.
- **Multi-User** – Linux is a multiuser system means multiple users can access system resources like memory/ ram/ application programs at same time.
- **Multiprogramming** – Linux is a multiprogramming system means multiple applications can run at same time.
- **Hierarchical File System** – Linux provides a standard file structure in which system files/ user files are arranged.
- **Shell** – Linux provides a special interpreter program which can be used to execute commands of the operating system. It can be used to do various types of operations, call application programs etc.
- **Security** – Linux provides user security using authentication features like password protection/ controlled access to specific files/ encryption of data.

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Topic: 5 Explain installation and configuration of Linux using ubuntu. Mark 5

Detailed:

Installation & configuration of Linux

Step 1 – Boot from the CD-ROM

- Installation of UBUNTU requires an installation (bootable) CDROM.
- To make CDROM as first boot device.
- When computer boots up it shows the options to enter the BIOS Setup. Generally, F2 or delete key is pressed to enter the CMOS (BIOS) setup.
- As CDROM is the first boot device, the computer will first boot from the CDROM and it would find the installation CD in the CDROM, and it will ask the user whether computer should be booted using CDROM. Press enters when it is asked to boot from CDROM.
- The installation process starts, and it initially greets the user with a welcome screen. For default installation, press Enter and for server version type server and press enter.

Step 2 – Selection of Language

- Ubuntu supports many languages, and this step in the installation is to select one of the languages from the given list. The default language is English.

Step 3 – Keyboard Layout

- Next, it asks to configure the keyboard layout. Generally, it will be automatically selected, so you can press Enter to ensure the selection. "Test your keyboard" option allows checking whether it is properly configured.

Step 4 – Hardware and Networking Auto Discovery

- Next, Ubuntu configure your hardware and network connection at this stage.
- If there is any network related error then we have option for "Do not configure the network at this time", so installation will not be interrupted due to this error.

Step 5 – Set a Hostname

- Installation asks for setting a host name for the computer.

Step 6 – Partition the Disk

- Partition has four option:

1. Resize IDEI master (IDE) and use free space
   Erase entire disk: IDE Master
1. Erase entire disk using LVM
2. Use target continuous free space
3. Manually edit partition table

- The Users can partition the disk manually by providing the appropriate space for the partitions.

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Step 7 – Wait during Automated package copying

Step 8 – Set the Time Zone

- Depending on your location, set the time zone for your locality.

Step 9 – Set your Real Name, Username and Password

- To login to the system, users require username and password, which needs to be unique.

Step 10 – Install the GRUB Boot Loader

- If windows is also installed on the same hard disk as Ubuntu, It will verify that GRUB boot loader is to be installed.
- The GRUB boot loader displays the list of operating systems installed on the computer when computer starts booting.

Step 11 – Reboot for the Second Installation Phase

Step 12 – Wait During Package Installation

Step 13 – Configure your Monitor

- After package are installed, set the resolution for monitor

Step 14 – Boot for the first time

- After rebooting it display the Ubuntu login screen, where user can enter the password and enter the system.

Topic: 6 Explain startup, shutdown, and boot loader in Linux. Mark 3

Ans.
Detailed:

shutdown

- If you are going to initiate a shutdown of your system and there are other users logged in to your system, it is always polite to notify them first.
- To do this, we would use the **shutdown** command. The shutdown command will accept times and a broadcast message:

  shutdown [options] time [warning_message]
- shutdown and reboot immediately

  $ shutdown -r now

Linux's boot process

- The Linux boot process is the name given to the startup procedures/order that your system goes through to load its operating system.
- In the following example, I will be referring to booting Linux on x86 architecture.

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Boot Loader

- The Ubuntu installation routine writes a new boot sector (also known as boot loader).
- The boot loader is a separate program called GRUB (Grand Unified Boot Loader).
- If more than one operating system is installed, GRUB boot loader with list of all installed operating system appears.
- The first option will be selected automatically within 10 seconds.

- In the boot loader screen, all pre-installed operating systems are listed, apart from that an entry ending in "(recovery mode)".
  This is a safe mode within windows, in that the system boots with safe system settings.

UNIT – 5 (Part-2)

Topic: 1 Explain Linux booting process. Mark 3

Ans.

Detailed:

- The linux booting process is series of 6 stages which is also known as start-up sequence.
- The booting process is initiated and completed during the time you press the power button and until the login screen appears.

1. BIOS

- BIOS stands for Basic Input/Output System
- Performs some system integrity checks
- Searches, loads, and executes the boot loader program.

2. MBR

- MBR stands for Master Boot Record.
- It is located in the 1st sector of the bootable disk.
- MBR is less than 512 bytes in size.
- So, in simple terms MBR loads and executes the GRUB boot loader.

3. GRUB

- GRUB stands for Grand Unified Bootloader.
- If you have multiple kernel images installed on your system, you can choose which one to be executed.

4. Kernel

- Mounts the root file system as specified in the "root=" in grub.conf
- Kernel executes the /sbin/init program

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5. Init

- Looks at the /etc/inittab file to decide the Linux run level.
- Following are the available run levels
    - 0 – halt
    - 1 – Single user mode
    - 2 – Multiuser, without NFS
    - 3 – Full multiuser mode
    - 4 – unused
    - 5 – X11
    - 6 – reboot

6. Run level programs

- When the Linux system is booting up, you might see various services getting started.
- For example, it might say "starting send mail …. OK".

Those are the run level programs, executed from the run level directory as defined by your run level.

Topic: 2 Explain LILO Configuration. Mark 3

Ans.

Detailed:

- LILO (Linux Loader) is a boot loader for Linux and was the default boot loader for most Linux distributions.
- Today, many distributions use GRUB as the default boot loader, but LILO is still in wide use.
- To work with LILO, the administrator needs to edit the file /etc/lilo.conf to set a default partition to boot, the time-out value, where LILO should be installed and other information.
- After that the administrator must update the loader by running the LILO command.

LILO Configuration File

- The following is a sample /etc/lilo.conf

```
boot=/dev/hda
Map=/boot/map
install=/boot/boot.b
prompt
timeout=50
message=/boot/message
lba32
default=linux
```

Topic: 3 Explain GRUB Configuration. Mark 3

Ans.

Detailed:

- GRUB is a boot loader designed to boot a wide range of operating systems from a wide range of filesystems.
- GRUB is becoming popular due to the increasing number of possible root filesystems that Linux can reside upon.
- GRUB is documented in a GNU info file.
- The **GRUB** boot loader uses the configuration file /boot/grub/grub.conf.

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GRUB Configuration File

```
default=0
timeout=10
splashimage=(hd0,0)/grub/splash.xpm.gz
```

- The 'default =' option tells GRUB which image to boot by default after the timeout period.
- The 'splashimage' option specifies the location of the image for use as the background for the GRUB GUI.

Topic: 4 Explain user interface GUI and CUI. Mark 3

Ans.

Detailed:

- User Interface is the junction between a user and a computer program.
- An interface is a set of commands or menus through which a user communicates with a program.
- A command-driven interface is one in which you enter commands.
- A menu-driven interface is one in which you select command choices from various menus displayed on the screen.
- The user interface is one of the most important parts of any program because it determines how easily you can make the program do what you want.
- A powerful program with a poorly designed user interface has little value.
- Graphical user interfaces (GUIs) that use windows, icons, and pop-up menus have become standard on personal computers.

GUI

- A GUI is a graphical (rather than purely textual) user interface to a computer.
- The term came into existence because the first interactive user interfaces to computers were not graphical; they were text-and-keyboard oriented and usually consisted of commands you had to remember and computer responses that were infamously brief.
- The command interface of the DOS operating system (which you can still get to from your Windows operating system) is an example of the typical user-computer interface before GUIs arrived.
- Today's major operating systems provide a graphical user interface.
- Applications typically use the elements of the GUI that come with the operating system and add their own graphical user interface elements and ideas.
- It is very user-friendly interface. Example: Windows, Linux

CUI

- CUI stands for Character User Interface of Command User Interface.
- In CUI user must interact with the application by making use of commands, in CUI only one task can run at a time.
- Everything is done by using commands, Example: DOS, UNIX

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UNIT – 5 (Part - 3)

Topic: 1 Explain how to create Linux user account and password. Mark 3

Ans.

Detailed:

Creating Linux User Account and Password on Ubuntu
To create a new user in Ubuntu, use the adduser command. The following is the syntax to create.

sudo adduser newusername

- The sudo is a program for computer operating system such as UNIX, Linux that allows users to run programs with the security privileges of another user, by default the Superuser.
- The name is combination of "Do as Su" where "Su" is for "Super user".

After typing that command following process started by Ubuntu:

- Type and confirm password.
- Enter in the user's information. This is optional. Press enter key to automatically fill the field with the default information.
- Press Y (or Enter key) when asked to confirm the information.

- You have just added a new user to your Ubuntu operating system.

Deleting a User

- If any time you want to delete a specific user on the virtual private server, you can delete them with the following command:

sudo user del username

- You should also delete the user's home directory. This can be done by following simple command :

sudo rm -rf /home/username

Topic: 2 Explain Samba server in detail. Mark 3

Ans.

Detailed:

- Samba is a software package, released in 1992, gives network administrators flexibility and freedom in terms of setup, configuration and choice of systems and equipment.
- Because of its ease and flexibility, Samba has become extremely popular and continues to do so.
- Samba runs on UNIX platforms, but can communicate to Windows clients just like the native platform.
- It allows a Unix system to move into a Windows Network without causing any mix-up.

- Windows users can easily access file and print services without having any problem.

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Installing and Managing Samba Server

- If you want to share files between your Ubuntu and Windows computers, the best option is the Samba.
- Download latest version of samba from the site: https://www.samba.org/samba/ftp/samba-latest.tar.gz
- To install, first open a terminal window and enter the following command:
  sudo apt-get install samba smbfs

- After the successful execution of this command, the Samba will get installed, now you must configure it to make it accessible.
- To configure it, open the configuration file using the following command in an editor:
  sudo gedit /etc/samba/smb.conf

  Find the following section in the file:
  # security = user

- Remove the comment before the security line and add the following line just below it:
  security = user
  username map = /etc/samba/smbusers

- This will set Samba to use the smbusers file for looking up the user list.

Creating Samba User

- There are two steps to creating a user.
- First we will run the smbpasswd utility to create a samba passwd for the user.

  sudo smbpasswd -a <username>

- Next, we will add the username to the smbusers file.
  sudo gedit /etc/samba/smbusers

- Add the following line, replacing the username with the one you want to give access to.
- The format is

<UbuntuUserName> = "<SambaUserName>".

Topic: 3 Explain Apache server in detail. Mark 3

Ans.

Detailed:

- The Apache Web Server is one of the most popular open-source HTTP servers.
- It is powerful, secure, and fully featured web server.

Installing Apache HTTP Web Server

- First, you should have the Apache server downloaded or download the Apache Server for Linux.
- Now extract the Apache Files by following commands:

gunzip -d httpd-2_0_NN.tar.gz
tar xvf httpd-2_0_NN.tar

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- To install Apache server on Ubuntu, use following commands:

  sudo apt-get update
  sudo apt-get install apache2

- The Apache server is installed.
- To check if Apache is installed, go to your browser, enter your server's IP address.
- The page should display the message: "It Works!"

Managing Apache Server

- You must understand the main configuration files of Apache which can be found in the "/etc/apache2" folder.

You can explore this folder by following command:

cd /etc/apache2
ls

you will see file listing. Following configuration files are necessary to understand:

apache2.conf : This is the main configuration file for the server. You can do almost all configurations within this file.
ports.conf : You can specify the ports from where virtual hosts can listen.
conf.d/ (Directory) : This directory is used for controlling specific features of the Apache configuration. For example, It is frequently used to define SSL configuration.
sites-available/ (Directory) : This directory contains all of the virtual hosts files that define different web sites.
sites-enabled/ (Directory) : This directory defines which virtual host definitions are actually being used.

mods-[enabled,available]/ (Directory) : These directories are similar in function to sites directories, but they define modules that can be loaded optionally.

Topic: 4 Explain how to optimize LDAP Services. Mark 3

Ans.

Detailed:

Optimizing LDAP Services

- LDAP (Lightweight Directory Access Protocol) is a protocol for managing information from a centralized location over the use of a file and directory hierarchy.
- It can be used to organize and store any kind of information.
- LDAP is mainly used for centralized authentication.

Install LDAP

- The OpenLDAP server is in Ubuntu's default package "slapd", so we can install it easily with apt-get.

- We will also install some additional utilities.

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sudo apt-get update
sudo apt-get install slapdldap-utils

You must enter administrator password for the admin LDAP account.

Reconfigure slapd

- When the installation is complete, we need to reconfigure the LDAP package.
  Type the following to reconfigure:

sudodpkg-reconfigure slapd

You must answer some questions about the configuration such as:
OpenLDAP Server configuration? Reply No.
DNS Domain Name? test.com
And so on…

Topic: 5 Explain DNS optimizing. Mark 2

Ans.

Detailed:

- We can also optimize DNS (Domain Name System) services on Linux (Ubuntu).
- The DNS is, as the name suggests, a simple naming system for your server so that you can remember the address by its name rather than its IP address.

Topic: 6 Explain FTP optimizing. Mark 2

Ans.

Detailed:

- FTP servers are a common way of sharing files between individuals with or without authentication.
- FTP is superior to HTTP (web) if the goal is to transfer files, not display information.
- After successful setup of an FTP server, you may need to optimize its services to get the better performance every time.
- For this, you need its proper configuration.
- FTP is file transfer protocol; security must be for safe file transfer.

Topic: 7 Explain how to optimize web services. Mark 2

Ans.

Detailed:

- Apache is a powerful and capable web server.
- To make initial setup as easy as possible, this makes it a great choice for new projects when you need to be productive quickly.
- Here are the few steps to optimize the web service performance of Apache Web Server.

1. Unload Unneeded Modules

- In Ubuntu based system, you will see a folder called /etc/apache2/mods-enabled and /etc/apache2/mods-available/.
- The mods-available folder is a list of all the modules installed on a particular server.

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- You should disable following modules if you do not need them:
  PHP
  SSL
  Rewrite
  Perl
  Python
  Rack / Ruby / Passenger

  2. Limit the Number of Apache Processes.

Topic: 8 Explain how to configure firewall in ubuntu. Mark 3

Ans.

Detailed:

- Firewall is considered as the first method of defense in securing your cloud server.
- Ubuntu includes its own (built-in) Firewall known as UFW.
- UFW stands for "Uncomplicated Firewall".
- Its main goal is to provide an easy-to use interface.
- The firewall is disabled by default. To enable the firewall, run the following commands from a terminal.

sudo ufw enable

- To install Firewall:

sudo aptitude install ufw
Or
sudo apt-get install ufw

- To check the status, use following command:

sudoufw status

Setup rules

- You should first define some rules for the firewall for allowing and denying connection.
- To set the defaults used by UFW, use the following commands

sudo ufw default deny incoming
And

sudo ufw default allow outgoing

- If you want to deny the outgoing request also, use the following command:

sudo ufw default deny outgoing

- Enable the SSH connection, use this command:

sudo ufw allow ssh

Deleting rules

- There are two option to delete rules.

sudo ufw delete allow ssh

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- If you need to reset your cloud server's rules to their default settings, you can do this :

sudo ufw reset

Topic: 9 Explain WINE. Mark 3

Ans.

Detailed:

- As we know that Linux is a great operating system, If you like a Window game or other app, you need to use Wine to run it right on your Ubuntu desktop.

Installing Wine

- You will find the Wine download in the Ubunto Software Center in all version.
- Download and install it.

Running an Application

- After successful installation of Wine, you can download an application's EXE file to install it by simply double-clicking it.
- You can also use terminal to install an application
- Use following command:

wine /path/to/application.exe

- After installation of the application, you can find it's shortcut icon on Desktop as we find it on Windows OS.

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